

1. Write a shell script that asks for the values of 2 variables: `userv1` and `userv2`, from the user and takes in any type (real/integer/character). Try to perform the following operations on the variables: (a) add, (b) multiply (c) subtract (d) division. Print appropriate error message/s if the operation cannot be performed, otherwise print the result: “the sum of ‘userv1’ and ‘userv2’ is ...”. Execute the shell script as many number of times the user wants. Show all possible combinations of the types of variables and result/s of the corresponding operation/s.
2. Write a shell script which accepts basic salary, House Rent allowance and Dearness Allowance of an employee and calculates net salary and displays the salary slip containing the Name of the Employee and the net salary.
3. Write a shell script which accepts a four digit number (integer) and a) displays the sum of its digits, b) displays the integer in reverse [that is, if you enter 7890, the program output will be 0987].
5. Write a shell script a) to find out the total count of files and directories in the current /any other directory. b) Also list the file and directory names separately with the respective counts. c) b) The script should also print out the total size of the files created in the past week.
6. Write a script that adds up the sizes reported by the `ls` command for the files in the current / any other directory. a) The script should print out only the total number of bytes used. b) The script should also print out the total size of the files created in the day that you are using this script.
7. The `PATH` variable is an environment variable that contains an ordered list of paths that Linux will search for executables when running a command. Write a shell script to display all the directories in the `PATH` variable in a simple way, i.e., one line per directory. In addition, display information about each directory, such as the permissions and the modification times.
8. Write a shell script which displays vendor id, model name, cpu MHz, cache size information about the processor present in your computer. [Hint: most of this information can be obtained by reading the file `/proc/cpuinfo`.]
9. Write a shell script *multiplication_table* that will generate a multiplication table for a number given on the command line.
10. Write a shell script to count the number of lines in a file.